

# Smart Water Filter Base

Design Document

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Group 8

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# Executive Summary

This design document covers the creation of our Smart Water Filter Base, which is a device that monitors the weight of water in a pitcher to determine the remaining water level and notify users when it is low.

Shared environments, including roommates, families, and offices, often rely on a single water filter. This can often lead to frustration if users frequently discover that the container is empty when they try to use it. This causes inconvenience, wasted time, and frustration.

Our proposed solution is a smart base that is placed under the filter and continuously measures the weight. This weight is converted into a percentage level and sent to users through a mobile app, notifying them if it gets low.

The system is composed of hardware, software, and web components.

## Hardware:

- Load cell for measuring weight
- Amplifier for sensor communication
- WiFi-enabled microcontroller for processing and communication
- Lithium battery for long term power

## Software:

- Microcontroller firmware for reading sensor data and sending notifications
- HTTPS server for communication between the microcontroller and the mobile app
- Website for initial setup instructions and account management
- Mobile app for displaying water level, group, and notifications

The Smart Water Filter Base is a simple, practical solution for monitoring water filter levels. It is easy to use and install, working with existing water filter pitchers without requiring any modifications. By providing real-time updates on water levels, it helps users avoid the inconvenience of discovering an empty pitcher when they need water.

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# 1 Introduction

## 1.1 Problem Statement

Living with roommates can sometimes mean your filtered water device is empty when you need water. This can be very frustrating and leave you needing to drink tap water while other roommates get to drink cold refrigerated and filtered water.

## 1.2 Need Statement

**We want a way to know if the container is out of water when we need water.**

- Avoid having no water left in addition to everybody using the water filter, not knowing there is no water left.
- Need a way to know the current water level of the pitcher so that it isn't empty without everyone knowing it's empty.
- We need to know if the container is empty, in order to not run out of clean fresh water.

## 1.3 Goal Statement

- Keeping water level known at all times, while informing everybody in the household if the water level runs too low.
- Create a system to inform users if the container is empty.

**If the container is empty, inform users.**

## 1.4 Personas and Users

**This is intended for shared households that rely on a shared filtration water container.**

- College Roommates
  - John Doe, 20 years old, Student at UCSC. John lives in a dorm with two other roommates, and they all rely on a shared water filter for drinking water. Occasionally, John comes out to get water only to find that the water is empty. This creates frustration and delays as John has to choose between missing the bus to refill water or getting to class on time while being thirsty.
- Family
  - Jane Smith, 39 years old, working mother. Jane lives with her husband and two children. Jane gets home after a long, exhausting day of work and is feeling thirsty from the drive home. She goes to the shared water filter to find that it is completely empty. Jane feels frustrated with her stay-at-home husband and two children for not refilling the water filter. This causes her to lash out and yell at them until they all cry, creating a hostile environment in the household.

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- Office

- Joe Micheal, 30 years old, works with Excel spreadsheets. He has very limited time to refill his mug, and his boss, George, never refills the water filter. As a result, Joe’s efficiency has substantially decreased at his desk.

## 1.5 Research into Existing Designs and Products

To understand where our design fits, we researched three existing products that partially meet the same need.

### Withings Body Smart

The Withings Body Smart is a consumer Wi-Fi scale that measures weight, body fat, muscle mass, and water percentage using strain gauge load cells and bioelectrical impedance analysis. It syncs data automatically to the Withings smartphone app, where users can track trends and set goals. While it offers a polished user experience, the system is entirely closed—users must rely on Withings’ proprietary app and cloud service, with no way to self-host data or customize the interface.

### Wyze Scale Ultra

The Wyze Scale Ultra takes a similar approach but adds a 4.3-inch color display directly on the scale, so users can view 13 body composition metrics without needing their phone nearby. It supports Wi-Fi syncing to the Wyze app and integrates with third-party fitness platforms like Apple Health and Google Fit. Like the Withings, it is a closed system that cannot be modified or repurposed beyond its intended body composition use case.

### Hackster.io ESP32 MQTT Weight System

On the DIY side, a Hackster.io project by The Embedded Things implements an ESP32-based weight measurement system using an HX711 amplifier and load cell. It features MQTT communication for real-time data streaming, multi-sample averaging for noise reduction, and automatic tare on startup. This project demonstrates strong signal processing but requires an external MQTT broker and separate dashboard software, adding complexity that our self-hosted approach avoids.

### Where Our Design Fits

Unlike commercial smart scales, our product is specifically designed for frequent monitoring of the fill level of a water pitcher. The system emphasizes event-based functionality, such as notifying the users of change in water level and when it needs to be refilled. Unlike the Hackster.io project, which focuses on the scale functionality and data transmission, our design features an end-to-end system including measurement, data storage, a web server, and a mobile app interface. A mobile app is much more accessible and easy to use than fully self-hosted or local solutions.

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## 1.6 Sustainability Statement

Our project promotes sustainable water consumption habits in shared households. Without knowing the current water level, users often open the fridge only to find an empty filter, leading them to either waste time waiting for it to refill or even use one-time plastic bottles. By providing real-time water level monitoring and low-level alerts, our system encourages refilling and reduces the likelihood of users turning to less sustainable alternatives. Over time, this helps households maintain consistent use of their reusable water filter rather than abandoning it out of frustration.

## 2 Design

### 2.1 Design Objective

Through the goal statement, it is needed that water level is known at all times to the user. To approach the goal, two parts needed being what hardware is needed to get accurate readings and what software is needed to communicate the information to the user.

#### 2.1.1 Hardware

We will be collecting data from the users through the weight of their water filters. All of the hardware will be contained in a plastic base which the filter will be set on. This will provide space for all of the internal electronics to be enclosed and kept dry in the humid refrigerator environment. The internal components will need to be sealed in completely to prevent them from getting wet.

The internal hardware components need to complete a few simple tasks. Detail about specific design choices will be explained in later sections focusing on specific component requirements. The following is a list of requirements for the hardware:

- Collect real time weight data
- Package weight data in an HTTPS packet and send it through Wi-Fi
- Work on battery power for long periods of time
- Charge the internal battery using a power port
- Have Wi-Fi and Bluetooth for initial setup and active use.

With these requirements in mind, our design uses a few internal components. Most components can be mounted to a custom designed circuit board. Others are connected with built in pins to the circuit board.

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### 2.1.2 Software

The requirements for the software design involve a few segments. They can be split into categories based on which components they interface with. Starting with the microcontroller code, this code needs to be capable of reading in values from the amplifier which provides serial data through an analog to digital converter. This serial data needs to be converted to a weight value and sent to the data collection server using a secure protocol. The microcontroller should also be capable of sending battery status values including a percentage estimate of the battery charge.

The software from the microcontroller will send packets containing info to the data collection server. This server must be capable of mapping incoming data to related user accounts that are subscribed to the specific device. Using a unique device ID and authentication token given by the server, incoming data can be assigned to the correct point. The device ID should be sent at the time of setup and will not change through operation. The server should keep tables containing users and their subscribed devices. This server needs to be secure and able to handle the traffic of a large number of devices communicating in real time.

Once the data is at the server, the mobile application used by users should get up to date information about the status of their filter and active fill value. This data should be updated close to real time in order for features such as notifications to work as intended. The mobile application must be capable of authenticating users using username and passwords unique to the user. The application must include information about each subscribed base, and have accurate information about their status. The app should include a notifications system that will alert users that the fill value has passed below the threshold. The application should communicate using secure protocols to prevent user data from being sent unsafely.

Notifications will be sent to everyone in the associated group when the water level drops below the threshold. This allows all users to be aware of the water level and give everyone the opportunity to refill it when it is empty, making it more likely that it will be refilled.

## 2.2 Physical Base Design

The physical base must consist of:

- A compartment containing all components (MCU, battery, load cell, amplifier)
- A place to rest a pitcher that can then be weighed
- A reset button to restart the device
- Humidity-proof seals that allow the device to operate inside the fridge environment

Our prototype implementation meets these requirements through the following specifications:

The physical base consists of two 3D-printed circular plates sandwiching a 75mm bar-type strain gauge load cell. Each plate is 110mm in diameter and 8mm thick, printed in PLA/PETG. The top plate has countersink pockets on the bottom face for M4 bolts that secure it to one end of the load cell, and the bottom plate attaches to the other end, sitting flat on the surface. The load cell wires route out to the amplifier and microcontroller on a nearby breadboard.

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### 2.2.1 Exploded View

Figure 1 shows the exploded view of the load cell assembly with all components labeled.

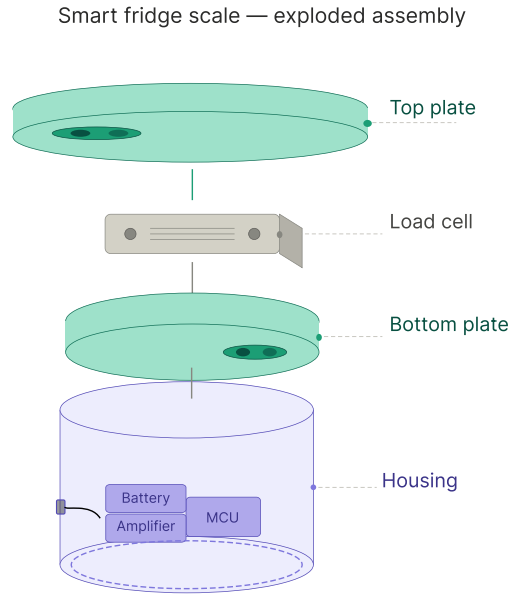


Figure 1: Exploded view of the load cell assembly showing top plate, bottom plate, load cell, and M4 hardware.

### 2.2.2 Base Plate Schematic

Figure 2 shows the design schematic of the base plate with dimensions and screw hole placement.

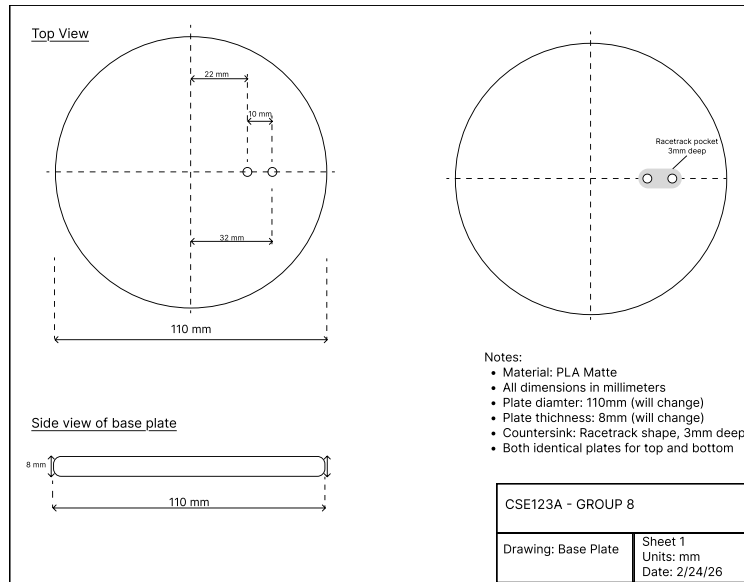


Figure 2: Design schematic of the base plate.

### 2.2.3 Rendering

Figure 3 shows how the complete system looks when assembled with a water filter pitcher placed on the scale platform.

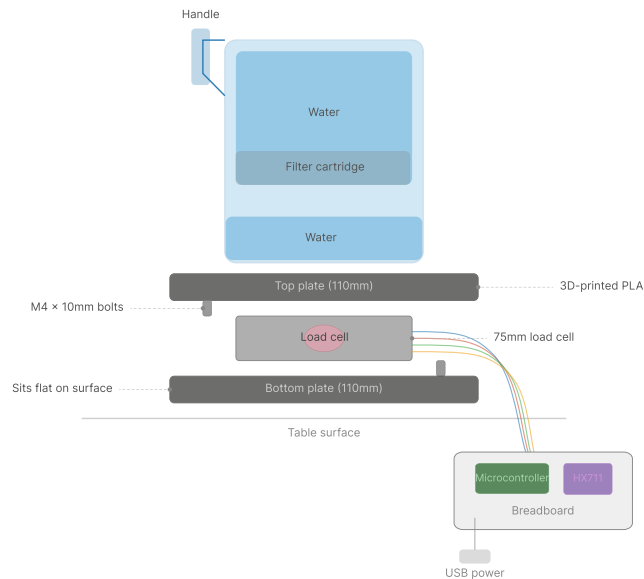


Figure 3: Diagram of the full system assembly showing the water filter pitcher resting on the load cell platform, with wiring to the microcontroller and amplifier on the breadboard.

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#### 2.2.4 Assembled View

Figure 4 shows the assembled base from a side view, with the load cell mounted between the two plates and wired to the breadboard.



Figure 4: Side view of the assembled base showing the two base plates, load cell, and wiring to the microcontroller.

#### 2.2.5 Completed Prototype

Figure 5 shows the completed smart base prototype.



Figure 5: Completed smart base prototype sealed.

### **2.2.6 Typical Use Case**

Figure 6 shows the finished smart base prototype in a typical use case.



Figure 6: Completed smart base prototype with a water filter pitcher on the scale.

## 2.3 Design for Manufacture

The design is intended for large scale manufacturing using a custom PCB. This custom PCB includes all required components such as the CPU, RAM, and Flash storage required to function as a microcontroller. It will also include power contacts for the battery, a power management unit for charging the included battery, and components like the WiFi antenna. The PCB design is an outline of what is required. Selected components can be changed and the circuitry altered as required if parts are unavailable.

The PCB will be mounted inside the 3D printed base and all other components will be contained within the base or related plates. The product is intended to be useable, and rechargeable without having to take apart or disconnect any internal components. This is so that it can be properly sealed from humidity to protect the sensitive electronic components.

The following diagram is the circuit diagram for the PCB we designed with currently available components. Additional components may be required if using different hardware than what we designed.

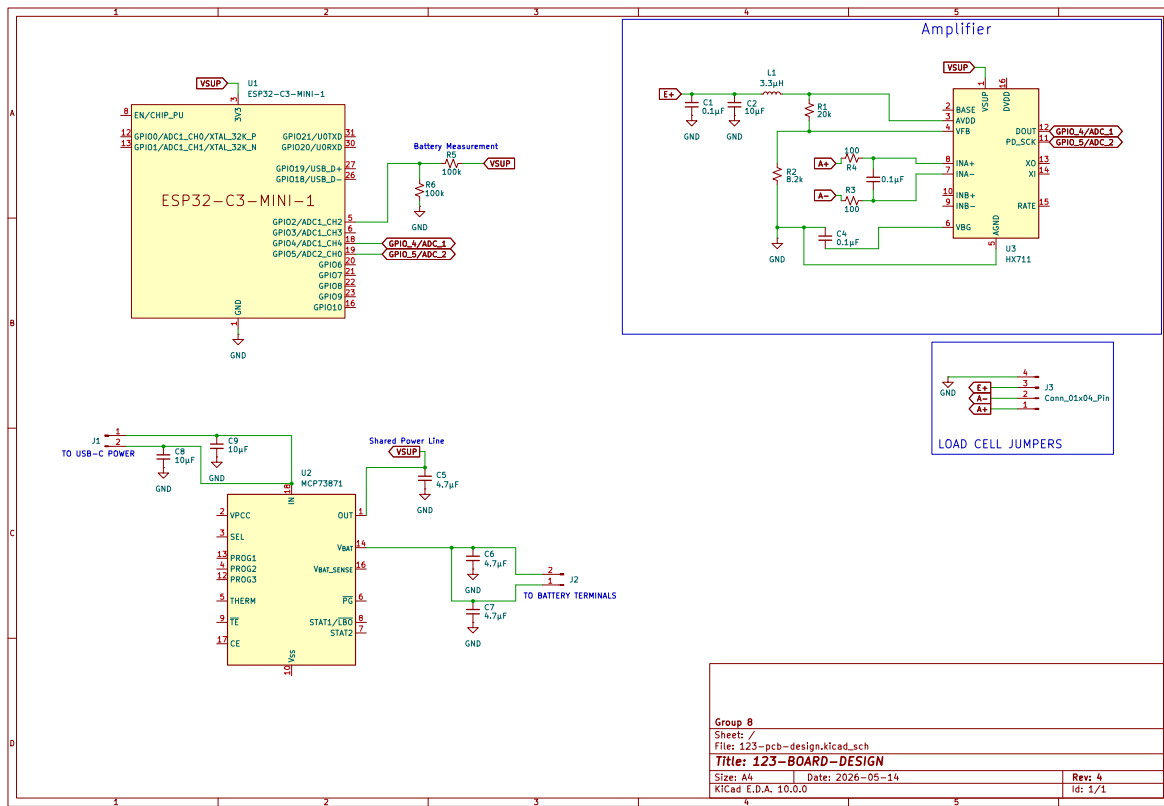


Figure 7: Wiring Diagram for PCB Components

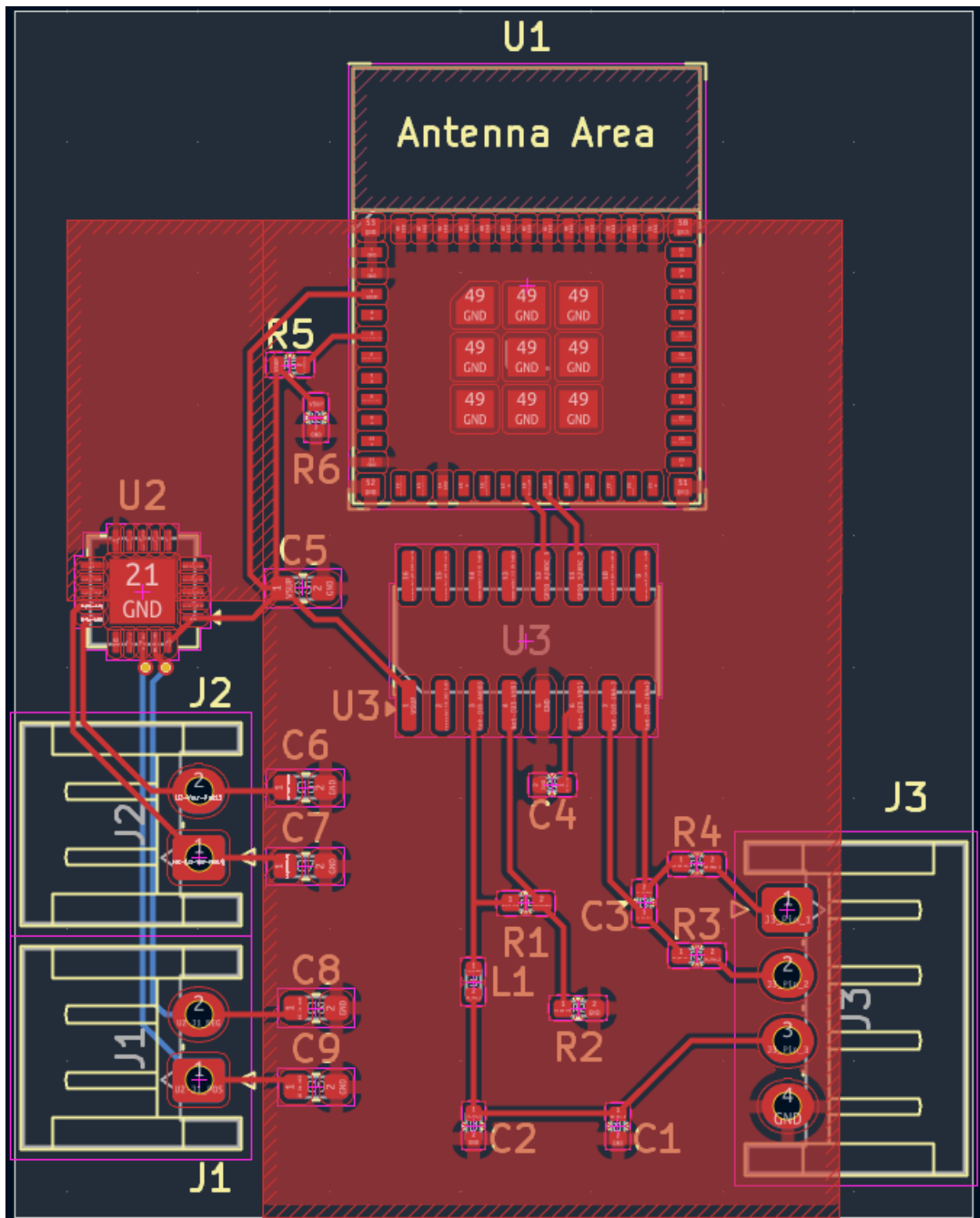


Figure 8: PCB Layout Using Commercial Components

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### 2.3.1 3D-Printed House

The base assembly consists of three 3D-printed parts: a top plate, a bottom plate, and a compartment at the bottom for the electronics. All parts are printed in PLA. The top and bottom plates are both 110mm in diameter and 8mm thick, with M4 screw holes and racetrack-shaped pockets for mounting the load cell. The plates went through multiple iterations to resolve flatness issues, which affected the stability of the load cell.

The compartment is a circular open-top tray with a 116mm outer diameter and a 110mm inner diameter, sized so the plates sit flush inside it. There is a resting shelf for the bottom plate to sit on, and tabs slightly above it so that when the plates are turned, they are held in place.

In the assembled stack-up, the compartment sits below the bottom plate so that the load cell remains free to flex under load while the electronics remain enclosed. Since the compartment is open at the top, the top plate is slightly wider, serving two purposes: to provide a cover for the compartment, and to be large enough for various sizes of pitchers to be placed on top of it.

In order to fully cover all components the base housing must be large enough to contain the load cell and associated electronics completely. For our design we chose to use two separate compartments within the base to separate out the electronics from the load cell. For a future design with the custom circuit board or a different load cell it could be possible to use a single compartment housing all components. When manufacturing at a large scale, using 3D printers may not be fast enough so the design could be changed to use a resin cast, or some other method of creating plastic components at a large scale for cheap.

### 2.3.2 Load Cell Assembly

Our design includes a load cell which has screw holes for attaching it to plates. Both plates are required and spacers are needed to bring the load cell off the plane, to a point where it is able to function as intended. This design is intended for a strain gauge load cell capable of measuring up to 20 kilograms. When designing for manufacture this part is a simple install with four screws. The cables required should be placed into a 4 pin connector so that it can interface properly with the custom PCB design. Using connectors rather than soldering will make the manufacturing process easier and reduce the chance that components break during shipping as these wires are very thin.

### 2.3.3 Electronics

The electronics will be contained inside a separate section of the 3D Printed base. The top section contains the load cell. The bottom section will have all of the mounting for the PCB, battery, and charging input. The wires for the selected load cell will come down through the bottom mounted plate through a small hole, then be connected to pins on the custom PCB. Wiring for the charging port will also mount to connector pins on the PCB. Wire length should be minimized to reduce the movement of internal components during shipping.

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#### **2.3.4 Power**

A battery is the best method of providing power for this product. In order to keep the user from having to repeatedly charge it and remove it from their fridge, a battery with a sufficiently large storage of milli-amp hours is necessary. Also included in the design is a method of charging the battery that does not require taking the device apart. The charging port will be able to charge the battery and keep the device powered so that users won't lose connection to their device.

#### **2.3.5 Amplifier**

The selected load cell for this design requires an amplifier in order to function properly. This amplifier powers the gauge and measures the return voltage through the resistor network. The design includes an amplifier capable of taking in microvolts and outputting 24-bit serial data. This serial data is interpreted by the microcontroller where it is turned to grams.

#### **2.3.6 Digital Converter**

To use the design, the software and hardware must be capable of interpreting analog data, converting it to digital data, and performing mathematical operations on the data to convert it to a unit of measurement. Our current design performs the interpretation and conversion to digital at the load cell. The digital data is sent to the microcontroller using GPIO pins where it is operated on by the formulas for conversion to grams.

#### **2.3.7 Average Sampling**

To reduce the chance of noise having a heavy affect on our readings, our design implemented an average sampling algorithm, where data is read in for a period of time, averaged, then interpreted as results. This not only prevents high spikes from noise, but also smooths the updating process, so a user placing or removing the filter rapidly will not have an unintended affect on the application, or cause issues like spamming notifications.

#### **2.3.8 Tare**

For handling offset, it is required to create a baseline state value that represents an empty pitcher. It means future data is can be compared to the baseline value to get the water weight readings. In our prototyping we found the offset value for our specific setup, but this value can change depending on the writing, pathway, and other factors. The offset value must be programmed into the device before is ready to ship.

## 2.4 Block Diagrams

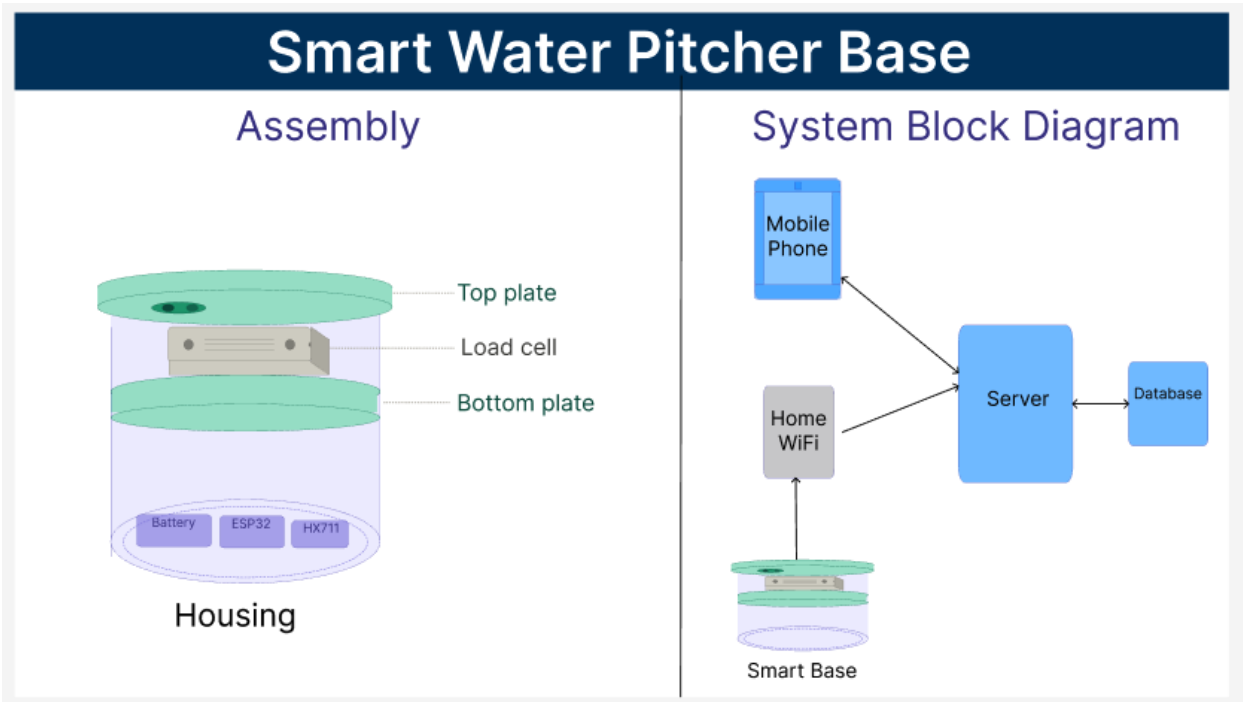


Figure 9: The system block diagram represents the pieces of the device and design, and how they are interconnected.

## 2.5 Wiring Diagrams

The wiring diagram for this design includes four pieces. The batter, microcontroller, load cell amplifier, and load cell. These parts all tie together with different wires connected to specific pins on each device. Each pin is labeled and each wire color coded. See Figure 7.

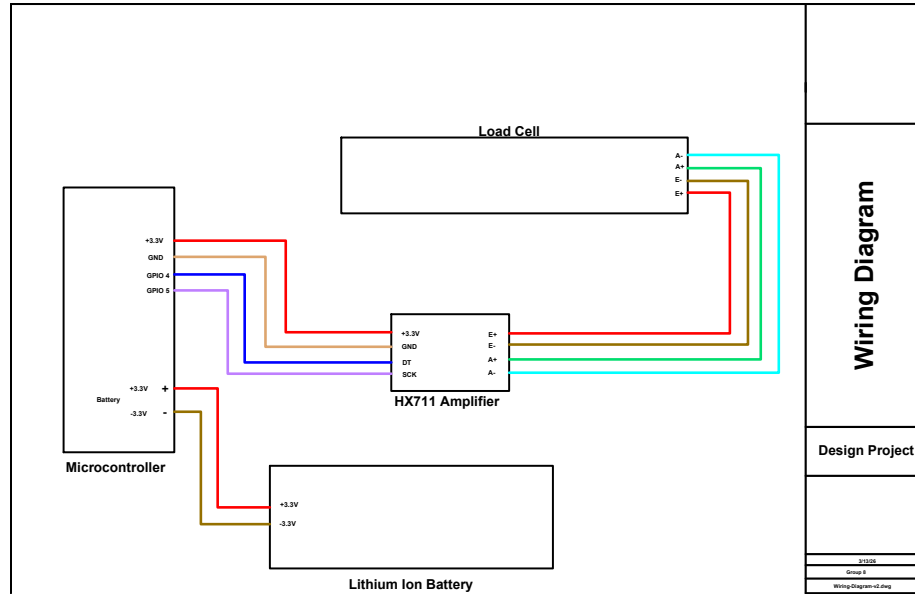


Figure 10: Example Wiring Diagram of Components Included in System Design

## 2.6 Prototype Implementation of Custom Design PCB

In order to produce large amounts of product, the cheapest and fastest method is to use a custom circuit board. This circuit board would ideally integrate all necessary components with as little extra work as possible. We provide an example prototype implementation using available components. These components can be swapped with other similar types, as long as their main functionality is still met.

The cost of this custom board will depend on the available components and their individual prices. With the design we have created, the estimated cost for the components of 1 board is \$9.99. This price does not include getting a bare board from a fab, or the cost to place the components. As production scales these costs will go down slightly as higher quantities of materials get a discounted per unit price. This board is designed to be simple to manufacture so that time is not spent in the build process.

### 2.6.1 Processor

The selected processor is the ESP32-C3-MINI-1. This processor is widely available and was selected for the following reasons. It includes 2.4GHz Wi-Fi and Bluetooth antennas in the packaging. This is ideal for our design as it reduces the number of required components. Adding a separate antenna is possible but would require a different chip selection that includes the required pins for Wi-Fi and Bluetooth.

The pins we use are the GPIO 4 and 5 pins. On this specific chip they are the ADC channels 0 and 4 respectively. These pins allow us to read data from our amplifier chip, and through the code we are able to interpret that as a weight value. These pins are required for this setup. Using another method like I2C or UART might be possible but would require a complete redesign to match our requirements. Analog to Digital pins are the simplest

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method for this design.

The power supply for the chip is provided through the 3.3v pin. We selected a power method that was capable of powering both the amplifier board and the ESP chip. Both can be powered with 3.3 volts. For most of their operational time, the ESP chip will be in standby mode which consumes very little power. The most power intensive time is when transmitting via WiFi as that has a tendency to spike the power consumption. In order to handle this, a power management circuit is required to prevent spikes in current.

Using a voltage divider to GPIO pin 2, a rough estimate of battery charge can be included in software. The board is set up to make this possible with the proper software implementation. By dividing down the voltage from the battery input, measurements can be made occasionally to determine the voltage output from the battery. This can be used with the knowledge that Lithium Ion batteries have a range in voltage output depending on their store of charge. The lower the output the lower the charge.

### **2.6.2 Power Components**

For this design we provide the MCP73871 as an example chip selection. This is a power management IC that is capable of switching between a USB or AC/DC power source, and a lithium ion battery. When the USB or AC/DC source is providing power, it will power the circuit load and charge the lithium ion battery. This chip selection includes a lot more features than our design takes advantage of, meaning room for expansion of features if desired.

The requirements a selected chip needs to include are the following. First, it needs to output the correct voltage for the selected amplifier and CPU combo. If not using an amplifier but some other method of data collection, then it needs to be able to output the required power for that method. Second, it needs to take two inputs, a USB or AC/DC input, and a lithium ion battery input. Third, it needs to be capable of charging the lithium ion battery, otherwise the product will only last a single charge and then not be useful to the customer.

Our selection for this design is to use a USB-C charging receptacle. This would allow us to place a floating port anywhere on the base that would be cheap to supply the cables to users. USB-C is a simple standard that can supply enough power if selecting a part that has only power pins enabled. The selected USB-C port in our designs contains the positive and negative power pins and nothing else. This meets the requirements for the selected power management chip. In order to use the design the way we intend you must select a USB-C or other power method that provides a positive and negative feed to the power chip.

### **2.6.3 Load Cell Amplifier**

The load cell amplifier is a required component for this specific design. Changing the design significantly to where it uses a different load cell than we had intended will require a redesign of this specific circuit component as it is only required for the load cell method selected. The load cell we describe requires a board which amplifies the microvolt output it provides. This chip selection is very important for processing the sensitive weight data produced by a load cell.

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The amplifier board must be powered from the same circuitry as the main chip to keep compatibility. In this design we specify 3.3 volts as the power source, but selecting a chip with a 5 volt or higher input would mean the selected amplifier chip must also be capable of 5 volt power, or wired up using a voltage divider circuit to get the required voltage.

The output of the amplifier board is used by the main CPU and program to determine the current weight placed on the sensor. In our design we specify that output going to analog to digital converter pins meaning the output of this device must be an analog signal, or something that can be interpreted by GPIO pins.

The inputs for this board come from four pins. We have designed the board so that it can be connected through a 4 pin JST connector. Any four pin connector that is small in form factor and can be soldered to the board horizontally, to lower the height requirements, is the best choice for this design. Keeping the height of the board minimal will mean the plastic base can be smaller and take up less space in a customers fridge. A 4 pin connector keeps the product from requiring soldering wires directly to a board, rather the wires for the load cell can be placed in a connector housing and plugged in directly. In order to seal the product well and prevent movement during shipping, a connector with a locking mechanism is the best choice for the design.

#### **2.6.4 Load Cell**

The load cell selection makes a big difference in how much the product can handle. For our prototype and our design we selected a 10 kilogram load cell which should be capable of weighing almost all commercial water filter pitchers. This selection means the most compatibility for the most consumers. Selecting a smaller load cell would limit the number of customers who can purchase it if they have larger pitchers capable of holding more than 4 liters of water. Even if the majority of consumers have smaller pitchers, we hope to provide the most availability.

The load cell in our design uses four wires to provide readings through a load cell amplifier. The design involves connecting the four wires of the load cell to the board using a 4 pin JST connector. The specific connector choice is described in the amplifier section. The boards will be physically separated by plates that have a small hole for cable routing. The cables should run from the load cell, through the hole, to the pin connection point on the custom PCB.

#### **2.6.5 Power Preservation**

The board will be operating through the use of a battery making it vital for the board to consume less power for longer operation duration. Developing a sleeping cycle to shut off components that do not be turned on all day. Wifi and Bluetooth require the most power in operation, making it necessary to have them on only when necessary, such as sending data or BLE provisioning. Since developing sleep and power, memory must be able to be preserved through sleep or sudden power shut offs to not lose data.

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## 2.7 Smart Base WiFi Provisioning

In order for the microcontroller to connect to the home network, the user must be able to send the WiFi credentials to the device. This is called WiFi provisioning. In our design, the user connects to the device through BLE and sends the data through the mobile app.

### 2.7.1 On Boot

When the device starts, it will attempt to load saved WiFi credentials and connect to the network. If this fails, it will enter provisioning mode. In our design, the MCU hosts a BLE access point that the user connects to through the mobile app. Once connected, the user will send the WiFi credentials through BLE to the device. The mobile app will also request a unique authentication token from the server for the device to use in the future, which will also be sent over BLE. The MCU will then attempt to connect to the specified WiFi network using the provided credentials. If the connection is successful, the device will save the wifi credentials to non-volatile storage and reboot.

### 2.7.2 Lost Connection

If WiFi connection is lost for any reason, the device will enter recovery mode. It will repeatedly attempt to reconnect, exponentially backing off over time to save resources. If connection is restored, the device will go back to normal functioning.

### 2.7.3 Manual Reset

If the user changes WiFi networks or needs to restart the device for any other reason, they can use the reboot button found on the smart base.

## 2.8 HTTPS Capabilities

The smart base must be able to securely communicate with a remote server or cloud service using HTTPS. This is necessary to send data about the water level, device identification data, and authentication data. It also must be able to receive authentication data or tokens from the server, which it will use for future communications.

The smart base must support:

- HTTPS client connections using TLS 1.2 or higher
- standard HTTP methods for bi-directional communication (e.g. GET, POST)
- server certificate validation to ensure secure communication
- reliable error handling for failed connections or invalid certificates
- ability to reestablish connections if the network is temporarily unavailable
- secure storage of any necessary credentials (e.g. API keys) for authenticating with remote servers

- 
- DNS resolution to connect to servers by hostname rather than IP address

The smart base must be able to securely transmit data to a remote server or cloud service for storage and analysis. This requires secure HTTPS client functionality, with the ability to both send and receive data. Additionally, the smart base should be able to handle common network issues or errors. Although DNS resolution is not strictly required depending on the implementation, it is strongly encouraged to allow for flexibility and scalability in the system design.

## 2.9 Web Application

The web application serves as the account-management and onboarding companion to the mobile app. It is built using React and deployed on a cloud platform, while being backed by a managed relational database. Sensor readings, group membership, device records, and authentication data are all managed within the backend infrastructure, which is shared between the web and mobile applications. The interface is organized into two side-by-side panels: an **Account** panel (the Settings page) and a **Setup guide** panel (the Guide page). The web app exists to let users manage the account that the mobile app signs in with, review the devices and groups associated with that account, and consult the first-time setup instructions.

### 2.9.1 System Architecture

The deployed architecture has three coordinated layers:

- **Frontend:** Renders the Settings panel and the Guide panel.
- **Backend Server:** Hosts the sensor-ingestion service used by the microcontroller and authenticated REST services used by both the web app and the mobile app to read and update account, group, and device data.
- **Database:** Provides authentication, a Postgres database for groups/devices/readings, and row-level-security enforcement.

### 2.9.2 Settings Page

The Settings page is the Account panel of the web app. It serves two states depending on whether the user is signed in.

**Signed out.** The panel renders a Sign-In form with email and password fields. A hint reminds the user to sign in with the same account they use in the mobile app, since the two clients share the same database authentication system.

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**Signed in.** After authentication, the panel displays:

- The signed-in user's email and a sign-out control.
- **My Groups** — a list of the groups the user belongs to, including the user's role in each group and the associated device ID where applicable. Group and device data are fetched from the authenticated REST services.
- **My Devices** — a list of devices owned by the user, showing each device's name and status.
- **Security** — a button to sign out of all devices currently signed in to the account and a button to delete the account. Deleting the account will revoke all devices and groups associated with the account.

All requests from the Settings page attach the database access token as a Bearer token; the REST services validate the token before returning data scoped to the user.

### 2.9.3 Setup Guide Page

The Guide page is a static, step-by-step walkthrough for first-time setup of a Smart Base. It is intended to be readable on a desktop screen while the user follows along with the mobile app. The guide covers:

1. Unboxing and powering on the base.
2. Installing the mobile app.
3. Signing in (or creating an account) and granting Bluetooth and Camera permissions.
4. Opening the Provision Device flow from the mobile dashboard.
5. Scanning the QR sticker on the base, which auto-fills the device name and ID.
6. Entering the Wi-Fi SSID and password the base should join.
7. Sending the authentication token and Wi-Fi credentials to the base over Bluetooth Low Energy.
8. Waiting for the base to confirm Wi-Fi connectivity and begin reporting readings.

The Guide page deliberately mirrors the mobile-app provisioning flow so users have a reference while completing setup on their phone.

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### 2.9.4 REST API

The backend server exposes a small REST API shared by the web app, the mobile app, and the microcontroller. User-facing requests carry the database access token as a Bearer credential; sensor uploads use a per-device credential established during provisioning.

- **Authentication** — Handles email/password sign-in, sign-up, and email verification, and issues access tokens for authenticated sessions. The web Settings sign-in form and the mobile Auth screen obtain tokens through the shared database authentication service; all other user-facing REST endpoints require a valid Bearer token before returning account, group, or device data.
- **Account and profile** — Read the signed-in user's profile, update display name, sign out all active sessions, and permanently delete the account. Used by the Settings page Security actions.
- **App state** — Returns aggregated state the clients need on load, including recent reading context for the signed-in user. Used by the Settings page and the Guide page.
- **Device registry** — List devices owned by the user and register a new device during mobile BLE provisioning (hardware identity plus one-time credential). Listing is used by the Settings page and mobile dashboard; registration is used by the mobile provisioning flow.
- **Group membership** — List the user's groups, create a new group, join an existing group by invite code, and fetch detailed group data including members and the latest reading. Listing is used by the Settings page and mobile dashboard; create, join, and detail are used primarily by the mobile app.
- **Group management** — Rename or delete a group, transfer ownership, and update member roles or remove members. Used by the mobile app.
- **Group calibration** — Set empty and full reference weights for a group's associated device. Used by the mobile Group screen.
- **Sensor ingestion** — Accept weight readings from the microcontroller over HTTPS, store them, derive water level from calibration data, and optionally trigger low-water push notifications. Used by the Smart Base firmware.

### 2.9.5 Data Model

The shared database stores the entities both clients rely on:

- **Devices** — Registered bases tied to a user account, including identity, display name, and provisioning credentials where applicable. Shown on the Settings page under My Devices and on the mobile app dashboard.

- 
- **Groups and membership** — Shared groups with invite codes, role-based membership, associated devices, and per-group calibration values. Shown on the Settings page under My Groups and on the mobile app dashboard.
  - **User profiles** — One record per user, including display name and authentication linkage. Used by both the web app and mobile app.
  - **Water readings** — Timestamped weight readings uploaded by the microcontroller. Consumed by the mobile app to show live water level.

## 2.10 Mobile Application

The mobile application is the primary user-facing interface for the Smart Water Filter Base. It is the client that displays live water-level readings for shared groups, handles device provisioning over Bluetooth Low Energy, manages device groups, and displays low-water notifications on the user's phone. The web application is limited to account management and the setup guide while the mobile app is where day-to-day interaction with the system happens. The mobile app shares its backend database with the web application. The same email/password account works in both applications, allowing users to manage their account and devices from both the web and mobile interfaces.

### 2.10.1 System Architecture

The mobile app interacts with the rest of the system in three ways:

- **Database Authentication** — Used for sign-in, sign-up, and live profile state. The same database authentication system is used for the web application and the mobile app.
- **Authenticated REST calls** — Sent to the backend server to fetch and update account, group, device, and calibration data, using the database access token as a Bearer credential.
- **Bluetooth Low Energy (BLE)** — For the initial device-provisioning handshake with a Smart Base. The BLE path is currently gated to iOS only.

Low-water notifications are scheduled by the mobile app itself and pushed directly to the user's phone.

### 2.10.2 Screens and User Flow

The mobile app is organized as a stack navigator with the following screens:

- **Intro Screen** — Shown on first launch. Introduces the product before the user can sign up and sign into the app.

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- **Auth Screen** — Email/password sign-in and sign-up, backed by the database authentication system. Sign-up optionally captures a display name. After account creation, the user receives a confirmation email and is required to verify their email before they can sign in.
  - **Dashboard Screen** — The home screen lists the user's groups and devices, lets the user create a new group, join an existing group by invite code, open the device-provisioning flow, and sign out.
  - **Provision Device Screen** — Walks the user through pairing a new base: scan the QR code on the base, enter Wi-Fi credentials, then transmit them to the base over BLE.
  - **Group Screen** — Detailed view for a single group. Displays the latest reading for the group's associated device, includes calibration controls for empty and full water levels, and is the destination opened when the user taps a low-water notification. Calibration is stored per group rather than as a single global record.

### 2.10.3 Device Provisioning

The Provision Device flow is the bridge between an out-of-the-box Smart Base and a cloud-connected device tied to the user's account. The flow proceeds in three stages:

1. **Scan QR** — The user scans a QR code on the physical base with the device camera. The app decodes the provisioning payload and pre-fills device identity and Bluetooth connection parameters for the next steps.
2. **Register on the server** — The app generates a one-time authentication token and registers the device with the backend so future sensor readings can be attributed to the user's account.
3. **Transmit over BLE** — The app connects to the base over Bluetooth Low Energy, sends the authentication token and Wi-Fi credentials, optionally reads a status response from the base, then polls the server until the newly registered device appears before returning the user to the Dashboard.

### 2.10.4 Groups and Sharing

A user's smart bases are accessed through groups. The Dashboard supports creating a new group (optionally associating it with a device the user owns) and joining an existing group via an invite code. Group owners have the option to remove members, transfer ownership, and delete the group.

The Group screen also allows groups owners to calibrate empty and full reference points. Calibration is stored per group (against the group's associated device) rather than as a single global record.

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### 2.10.5 Low-Water Notifications

The mobile app monitors its own group devices for low-water conditions and surfaces local notifications when the level drops below threshold. Three pieces work together:

- **Background monitoring** — While the user is signed in and the app is active, a background process periodically fetches the latest reading for each group that has an attached device.
- **Threshold detection** — The app computes each group's water-level percentage and compares it against a fixed low-water threshold. Per-group state ensures one notification per low-water event rather than repeated alerts while the level stays low.
- **Local notifications** — When a group crosses below threshold, the app schedules a local notification. Tapping it opens the corresponding Group screen.

## 2.11 Summary of Prototype Implemented Features

- **Smart Base**
  - **Implementation:**
    - \* ESP32-C3
    - \* 900 mAh Li-ion battery
    - \* 3D printed container
    - \* Load cell, HX711 amplifier
  - **Implemented features:**
    - \* Setup and connection with mobile application
    - \* Send weight through HTTPS
  - **Missing features:**
    - \* Reset button
    - \* Humidity proofing
- **Mobile App and Server**
  - **Implementation:**
    - \* React Native on Expo iOS
    - \* Vercel hosting, Supabase database
  - **Implemented features:**
    - \* Setup, add, and calibrate devices
    - \* Create/join/manage groups
    - \* View live water level
    - \* Receive low-water notifications
  - **Missing features:**
    - \* Customize water level threshold to notify
    - \* Full push notifications capabilities (locked by Apple paywall)

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## 3 Evaluation

### 3.1 Testing

Testing is a critical phase in the development of the smart water-level monitor, ensuring that all components function correctly and integrate seamlessly. We will conduct testing at multiple levels, including unit testing for individual modules, integration testing for combined components, and end-to-end testing for the entire system.

Our local tests for the MCU are linked inside the repository under `Code/Links to Test Cases.md`.

#### 3.1.1 Microcontroller

When creating the tests for the microcontroller code, it's important to consider what a user behavior might look like, but also important to understand what level the behavior could get to. How likely is it that a user is putting a full filter on, then an empty one, and back to full, in a matter of milliseconds? How capable is this controller of dealing with sharp spikes and changes? The testing plan covers both simulated behavior for quick and sharp event behavior, as well as a more practical testing plan for typical user behavior.

In order to verify the function of the code reading from the Wheatstone bridge sensor, we must verify that the logic itself works the way we think. There are multiple parts that require logical verification.

First, the sensor data will come in with some small variation depending on where the weight is placed on the base, and because ADC values can vary slightly without any other input. Sensor reading code must also account for the fact that data can change very quickly, so testing the logic through repeated, high frequency changes, can help show its robustness. There is also the issue of whether data inputted to the code is read in correctly through the bit shift operations, so checking the shift operations for correctness is important.

Testing the microcontroller code in this design involves a separate file that contains simulated amplifier behavior. This behavior is passed into the functions used on the microcontroller, which provide an output back to the test file. These outputs are verified against expected outputs and if they are the same or within a certain level of tolerance, then we can say the code passed the tests.

To test the behavior that would be seen for a typical user, we must have standard objects that we use for the different levels of weight. A large weight, a small weight, and no weight are the simplest form of this. For example, fitting with the project, say a water filter with 100mL of water, versus a water filter with 3L of water. These two objects have drastically different weights so putting it on the scale and viewing how the code responds could represent a typical user behavior of taking the empty filter, filling it, and returning it to the base.

#### 3.1.2 Mobile Application

When testing the mobile application, we focus on both normal use and edge cases. A normal flow is a user opening the app, calibrating once, and checking the water level. Edge cases include repeated page refreshes, pressing calibration buttons several times, or receiving rapid updates from the sensor. We also test edge cases like if the water level goes beyond expected

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limits. We want to ensure the app can handle these scenarios without crashing or showing incorrect information.

For frontend unit tests, we check important UI parts one at a time, such as calibration buttons, status messages, and notification prompts. These tests verify that components render correctly, user clicks update state as expected, and displayed values stay consistent with incoming data. Because sensor readings can change often, we also check that the UI updates smoothly and does not show stale values.

For integration testing, we verify that the frontend and backend communicate correctly. The app sends requests for actions like token registration and sensor data ingestion, and we check that the API returns the expected results. We test both valid and invalid payloads to make sure errors are handled clearly and data is not updated incorrectly.

Finally, end-to-end tests cover complete user workflows from start to finish. These include loading the app, calibrating, monitoring water-level changes, and receiving notifications at threshold events. If these tests pass under both normal and adverse conditions, we can conclude the mobile application meets functional testing goals.

## 3.2 System Testing

### WiFi Capability & Provisioning

**Objective:** Verify that the Smart Water Pitcher Base can be provisioned with WiFi credentials and reliably connect to a wireless network for communication with external devices.

**Test Cases:**

- **Initial Provisioning Test**
  - **Procedure:** Power on the device in provisioning mode and use the setup interface to enter WiFi SSID and password.
  - **Expected Result:** The device stores the credentials and connects to the specified network.
  - **Prototype Result:** Pass
- **Automatic Reconnection Test**
  - **Procedure:** Power cycle the device after provisioning has been completed.
  - **Expected Result:** The device reconnects automatically to the saved WiFi network without requiring reconfiguration.
  - **Prototype Result:** Pass
- **Invalid Credential Handling**
  - **Procedure:** Attempt provisioning using an incorrect WiFi password.
  - **Expected Result:** The device fails to connect and indicates a connection error or returns to provisioning mode.
  - **Prototype Result:** Pass

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- **Network Loss Recovery**

- **Procedure:** Disconnect the router or move the device out of range, then restore the network.
- **Expected Result:** The device automatically reconnects once the network becomes available.
- **Prototype Result:** Pass

**Success Criteria:** The device can be provisioned successfully, stores WiFi credentials correctly, and maintains reliable connectivity to the network during normal operation.

## Group Management Test Plan

**Objective:** Verify that users can join and leave a group associated with a specific Smart Water Pitcher Base and that group membership is updated correctly in the application.

### Test Cases:

- **Join Group Test**

- **Procedure:** A user enters the group code or selects the filter in the application to join its group.
- **Expected Result:** The user is successfully added to the group and gains access to the filter's information.
- **Prototype Result:** Pass

- **Leave Group Test**

- **Procedure:** A user selects the option to leave the group in the application.
- **Expected Result:** The user is removed from the group and no longer sees the filter's data.
- **Prototype Result:** Pass

- **Multiple User Membership Test**

- **Procedure:** Multiple users join the same filter group from different devices.
- **Expected Result:** All users are successfully added and can view the same filter information.
- **Prototype Result:** Pass

- **Invalid Group Handling**

- **Procedure:** Attempt to join a group using an invalid or non-existent group code.
- **Expected Result:** The application rejects the request and displays an error message.
- **Prototype Result:** Pass

- **Rename Group Test**

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- **Procedure:** Attempt to rename a group.
  - **Expected Result:** The group name is updated correctly for all users.
  - **Prototype Result:** Pass

- **Delete Group Test**

- **Procedure:** Attempt to delete a group.
- **Expected Result:** The group is removed for all users.
- **Prototype Result:** Pass

**Success Criteria:** Users can reliably join and leave groups associated with a filter, and group membership is accurately reflected across all devices.

## Water Level Monitoring and Display

**Objective:** Verify that the Smart Water Pitcher Base accurately measures the water level and correctly displays this information to users through the application.

### Test Cases:

- **Sensor Calibration Test**

- **Procedure:** Empty the container and select the “Empty” calibration option in the application. Then fill the container to the maximum level and select the “Full” calibration option.
- **Expected Result:** The system stores the calibration values and uses them to correctly scale future sensor readings to the displayed water level.
- **Prototype Result:** Pass

- **Water Level Accuracy Test**

- **Procedure:** Fill the container to several known levels and record the value reported by the Smart Water Pitcher Base.
- **Expected Result:** The reported level closely matches the actual water level within an acceptable error range.
- **Prototype Result:** Pass

- **Empty and Full Level Test**

- **Procedure:** Test the system with the container completely empty and completely full.
- **Expected Result:** The application correctly displays the minimum and maximum water levels.
- **Prototype Result:** Pass

- **Data Transmission Test**

- **Procedure:** Verify that sensor readings are transmitted from the Smart Water Filter Base to the application over WiFi.

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- **Expected Result:** The application consistently displays the most recent water level data.

- **Prototype Result:** Pass

- **Transmission Speed Test**

- **Procedure:** Verify that sensor readings are transmitted from the Smart Water Filter Base to the application within 5 seconds.
- **Expected Result:** The application consistently reports water level data within 5 seconds of a physical change.
- **Prototype Result:** Pass

**Success Criteria:** The Smart Water Pitcher Base accurately measures water level and reliably communicates this information to the user interface.

## Notification System Test Plan

**Objective:** Verify that the system generates and delivers notifications to users when specific conditions occur, such as low water levels or filter maintenance requirements.

### Test Cases:

- **Low Water Level Notification**

- **Procedure:** Reduce the water level below the defined threshold.
- **Expected Result:** A notification is generated and displayed to users indicating that the water level is low.
- **Prototype Result:** Pass

- **Notification Delivery Test**

- **Procedure:** Trigger a notification event and verify that it appears in the application interface.
- **Expected Result:** The notification is delivered and visible to the user.
- **Prototype Result:** Partial pass (Apple Developer Kit prevents full notification access without payment)

- **Multiple User Notification Test**

- **Procedure:** Trigger a notification event while multiple users are members of the same group.
- **Expected Result:** All users in the group receive the notification.
- **Prototype Result:** Pass

**Success Criteria:** Notifications are reliably generated and delivered to all relevant users when system conditions trigger an alert.

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## Miscellaneous Functionality Tests

### Test Cases:

- **Environmental Humidity Test**
  - **Procedure:** Place the Smart Water Pitcher Base in an environment with known humidity levels similar to a typical indoor refrigerator.
  - **Expected Result:** The smart base continues to function correctly over long periods of use.
  - **Prototype Result:** Untested, prototype is not humidity proof
- **Battery and Power Test**
  - **Procedure:** Drain the smart base's battery and recharge it.
  - **Expected Result:** The smart base resumes functionality after being recharged.
  - **Prototype Result:** Pass
- **Battery Duration Test**
  - **Procedure:** Measure how long the battery lasts under normal usage conditions.
  - **Expected Result:** The smart base remains powered for 3 months without recharge.
  - **Prototype Result:** Fail, prototype's MCU uses much more power than the actual design's custom PCB

## 3.3 Web Application Testing

The web application testing plan verifies both the user-facing interface and the backend API that the web app, mobile app, and microcontroller share. Tests run locally with mocked authentication, database responses, and network calls so behavior can be checked without live hardware or a production database. Frontend tests use a component test library to render pages and simulate user actions. The backend tests invoke API handlers directly and assert on status codes, response bodies, and error handling. All automated tests live under the `test/` directory in the web-server source tree and are executed with a single test runner command.

### 3.3.1 Frontend and User Interface

Frontend tests focus on the two panels: the Account Settings page and the Setup Guide page. These tests confirm that the application shell renders both sections, that unauthenticated users see a sign-in form, and that authenticated users see account information loaded from the API. For the Settings page, tests cover the full signed-out and signed-in flows. When no session exists, the page shows a loading state and then the sign-in form. Invalid credentials display an error message from the authentication service. After a successful sign-in, the page lists the user's groups and devices, handles API load failures gracefully, and updates when the session changes. Users can sign out, which clears displayed account data. Security

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actions are also verified: account deletion requires confirmation and only proceeds when the user accepts, and cancellation leaves the session intact. The Setup Guide page is tested at the layout level: both the Account and Setup Guide headings appear together so users can read setup instructions alongside account management. Because the guide is largely static content, testing emphasizes correct rendering rather than dynamic data flows.

### 3.3.2 Backend API and Authentication

The web server also hosts REST endpoints used by the mobile app and the microcontroller. A separate set of tests validates server-side logic in isolation. Each test calls an API handler with a mock request and inspects the HTTP response, using stand-in database and auth clients so tests do not depend on external services. Authentication tests verify that user requests without a valid Bearer token are rejected, that device ingestion requests require valid device credentials, and that invalid or replayed tokens are not accepted. These checks protect endpoints that read or write user and device data. Group-management tests cover listing, creating, joining, renaming, and deleting groups, as well as updating calibration values and managing members (including role changes and removal). Tests include both successful paths and failure cases such as invalid invite codes, duplicate membership, missing permissions, and database errors. Device tests verify listing a user's registered devices and the BLE registration flow used during mobile provisioning, including linking a new device to an existing group and handling missing or invalid input. The sensor ingestion endpoint is tested for correct handling of POST requests, device authentication, weight-to-level conversion using calibration data, storage of readings, and optional low-water push notifications. Profile endpoints are tested for reading and updating user information, signing out all active sessions, and deleting an account. Supporting utility modules, such as push-notification delivery and shared group logic, are also tested so edge cases (empty inputs, network failures, malformed responses) do not propagate into production handlers.

## 3.4 Testing the Amplifier

To test the code written for this project, we first had to create a simulated version of the sensor. Regardless of what the sensor reports we want to make sure that all of our math and interpretation of incoming data is correct. To do this, the functions were adapted to either take input from GPIO pins, or from a specific test file.

### 3.4.1 Test File

Inside the test file are adapted versions of each function used by the main HX711 file. These include the following:

- `hx711_set_sck`
- `hx711_get_dout`
- `read_raw`
- `get_raw_weight`

- 
- tare
  - get\_grams

With each of these functions, the test file simulates reading data from the sensor. This code was created to simulate our selected load cell and amplifier combo. The design includes this amplifier. The code will need to be adapted for any other amplifier that uses a different form of communication or a different packet structure. This code can be adapted by changing the functions specifically related to the amplifier labeled with HX711.

### 3.4.2 Simulated Data

Using some set values defined at the top of the file including a hardcoded offset and scaling factor, the test file first sets the weight and tares it. Then based on a random number between 0 and 2, it sends either a 0, small weight, or large weight value.

When the test file calls to the read weight functions, some small noise of  $\pm 1g$  is used to simulate the variability of the sensor reading. The sensor sends data in 24 bit samples, so the test code also includes the 24 bit shifting. Once the simulated data is created, the main function compares it to the expected result. If it's within one gram of the expected low high or zero value, then the value passes. This tolerance is arbitrary for the testing and will be different for the production code.

## A Problem Formulation

### A.1 Conceptualizations for Design Approach

#### A.1.1 Contactless Capacitive Water Level Sensing

A contactless sensor is a small puck that attaches to the side of the water filter. It would work by detecting when the level goes below a certain point, and sending that information to the microcontroller. It's a simple and easy way to do one level detection. It produces a binary output of either above or below the detected level.

#### Pros

- Low-cost
- No contact
- Small mechanical approach

#### Cons

- Need a good environment for sensor, not humid so not great inside a fridge
- Users need to install device themselves, making it hard to standardize what's considered "empty"
- Binary output doesn't allow for percentage based information, or any form of consumption metrics.

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Because of the noted cons to this design style, we decided to find another solution. The user needing to place the sensor on their own device could throw off any configuration we could complete on our side. It puts too much onto the user for it to work well.

### **A.1.2 Weight based, Load**

The design places the pitcher unto a load cell to measure the weight. We will calibrate it to the changes in weight.

#### **Pros**

- Highest accuracy of all designs
- Fits to many pitcher dimensions

#### **Cons**

- Need a good environment for sensor, not humid so not great inside a fridge

### **A.1.3 Ultrasonic**

An ultrasonic sensor estimates the water level by measuring the echo time of sound waves reflected from the water surface. The sensor would be installed attached to the side of the filter as close to the top of the water compartment.

#### **Pros**

- Inexpensive
- Easy implementation

#### **Cons**

- Humidity problem
- Mounting hardware and wiring sensor is difficult without modifications to container.

If there are modifications to the filter itself in order for a design to work, then the design could not be possible to produce. If we chose this design and needed to modify the pitcher to mount the hardware or get the wiring right, then we would have to sell pre-modified pitchers as well, otherwise users might not be able to use the device properly. Because of these reasons we decided against using this design style.

### **A.1.4 Laser**

A laser or time-of-flight sensor measures the distance from the sensor to the water surface. The measured distance is converted into liquid height and volume.

#### **Pros**

- Continuous measurements
- Non contact

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## Cons

- Humidity problem
- Clear line of sight
- Mounting problems

Some of the same reasons we didn't choose the ultrasonic sensor apply to this laser sensor design. On top of that in order for us to use continuous measurements, the battery power required would raise and we'd have to figure out how to power the device for long periods of time.

## A.2 Brainstorming

Items to decide on:

- Microcontroller
- Sensor
  - Ultrasonic: Pointed down from the lid onto a floating object
  - Load Cell: Place the container on the load cell to measure the weight and calculate the water level
  - Laser: Shoots a laser into the water from the lid, and the distance is sent back to the sensor
  - Camera: Setup to watch the water filter and measure the water level based on computer vision
  - Non-Contact Sensor: Patch that sits on the side, when water level goes below sensor it can alert the device
- Server
  - HTTP
  - AWS
  - Vercel
- Challenges
  - Signal in the fridge
  - Size limit
  - Contact Vs No Contact
  - Wifi connectivity
  - Power source (battery, USB, etc.)

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| Criteria (Weight)       | Capacitive | Weight      | Laser       | Ultrasonic  |
|-------------------------|------------|-------------|-------------|-------------|
| Cost (0.30)             | 5          | 5           | 3           | 5           |
| Power (0.20)            | 5          | 4           | 3           | 4           |
| Complexity (0.25)       | 3          | 4           | 2           | 3           |
| Measurement Data (0.25) | 3          | 5           | 3           | 3           |
| <b>Weighted Total</b>   | <b>4</b>   | <b>4.55</b> | <b>2.70</b> | <b>3.85</b> |

Table 1: Decision matrix comparing feasible sensing approaches (1 = poor, 5 = excellent)

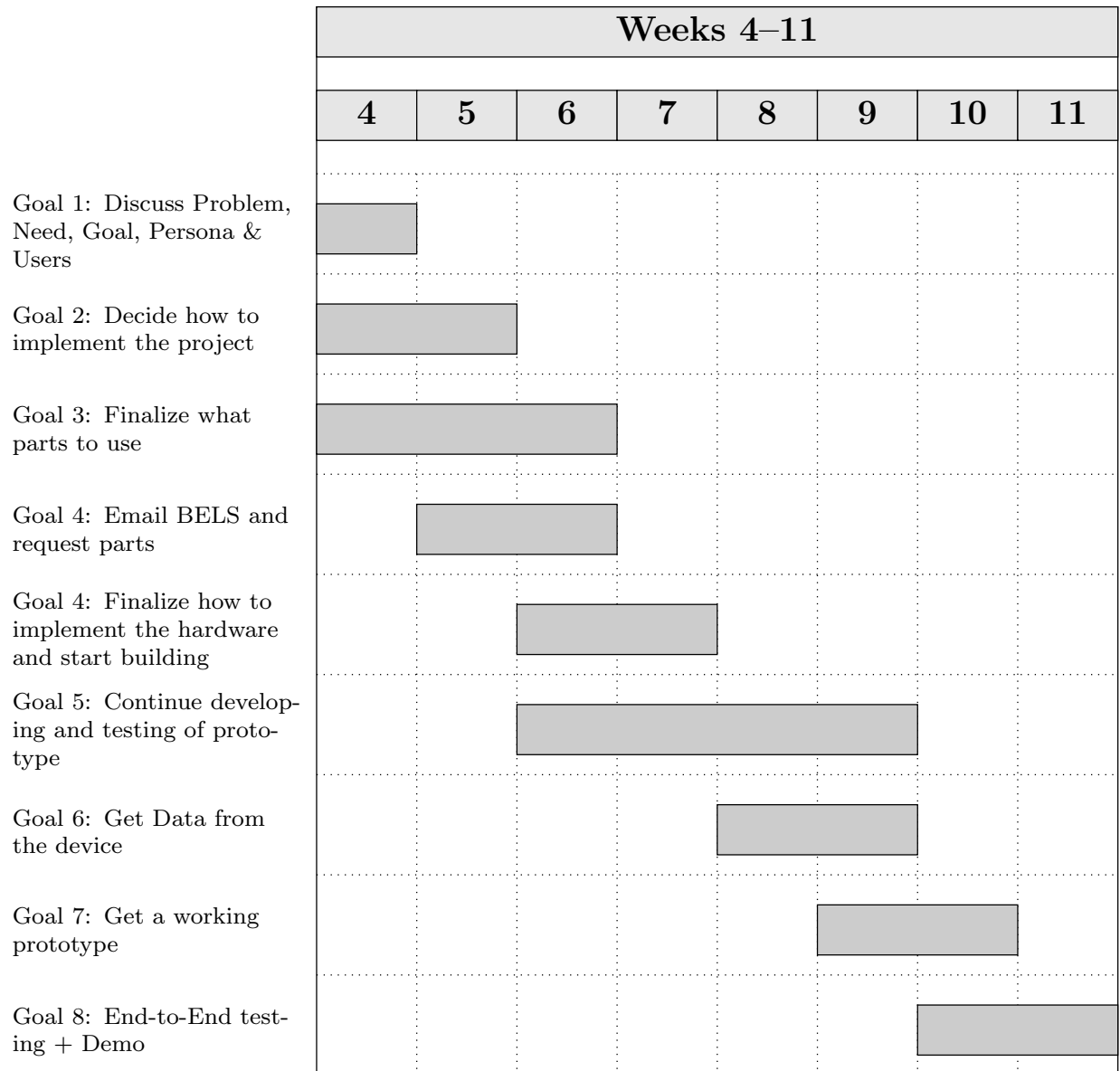
### A.3 Decision Table

Based on the decision matrix, the weight sensor is the best sensing method because it has the highest weighted total score (4.55). It performs especially well in measurement quality while still keeping cost, power, and overall complexity at a strong level compared to the other options.

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## B Planning

### B.1 Basic Plan / Gantt Chart



### B.2 Division of Labor

**Dev Chodavadiya:** For this quarter, I focused on the physical design, software features. For hardware, I designed and 3D-printed the housing for the product for the water filter base, iterating on multiple versions to achieve the correct height and correct locking mechanism. For the software, I worked on implementing the user signout feature and account deletion functionality. I also contributed to documentation by writing the design document sections on the physical design and software features, as well as some of the drawings and diagrams.

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**Logan Bossuwe:** I handled the Load Cell and HX711 sensor hardware to the microcontroller. I soldered the board with Ried that has the amplifier and microcontroller on it to be placed inside the prototype. Additionally, I printed the final iterations toward our aesthetic prototype. For the software, I implemented HX711 sensor code for accurate readings and calibrated it to output grams to be sent via HTTP. I created a main file where I combined other members code related to the microcontroller into it and ensured safety with memory depending on the boot cycles. For the main code file, I added a sleep cycle for power preservation, which had to system designed with the other parts such as BLE, HTTPS Post and NVS. For documentation, the main contribution was conceptualizing design ideas with a decision table and focused on proper documentation load cell amplifier and power preservation methods for the battery. I worked on early implementations of diagrams for the amplifier and microcontroller.

**Pieter Nauwelaerts:** My main contribution to this development was in the form of creating a custom board design to match the project requirements. I designed the circuitry and found all of the required components to implement our design properly. I used these components to design a board layout that could be manufactured as it passes all design rule checks. This board includes all of the features our microcontroller design has, including wifi/bluetooth, power management through a USB port or battery, and the load cell amplifier.

I also contributed to design document updates in making the document more professional and design focused for someone to try and recreate our design in the future. I created diagrams, images, and designs for the poster, presentation, and design document.

**Reid Graham:** I mainly worked on the physical smart base, documentation, and planning. For the smart base, I worked on the physical assembly of the device, including printing, soldering, and plate assembly. For the smart base software, I created an HTTP and HTTPS library, as well as multiple versions of libraries for WiFi connection, reconnection, and provisioning.

I also worked to plan basic feature capabilities as well as the test plans. I organized team meetings and communicate with the course staff about various issues. Since I could not easily work on the mobile application without a Mac device, I also contributed more to the poster, presentation, and design document, as well as created diagrams.

**Sam Lai:** I mainly worked on the web application development and testing. Specifically, I implemented the notification system that allows users to receive alerts on their laptops and phones when the water level of the water pitcher is low. I implemented two API endpoints to handle subscription management and notification sending. In terms of the design document, I contributed to writing and organizing the mobile and web application code and test sections. I also proposed the idea of using a weight sensor and wrote down brainstorming ideas for the design. In spring quarter, I mainly worked on developing the mobile app and writing tests for both the mobile and web applications. I created the login and signup screens, the dashboard screen, the group screen, and any functions within each of them. I also implemented the user login authentication system and the notification system for groups. I also implemented authenticated api endpoints in the backend server for the mobile app to inter-

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act with the backend server to query and update data from the database.

**Shriyan Gote:** I mainly worked on the web application frontend and deployment. I designed and implemented the Brita water level dashboard UI, including the pitcher graphic, animated water fill, percentage display, last-updated time, and calibration controls (calibrate empty/full) that store values in Supabase. I set up Supabase Realtime so the page updates when new sensor data is posted, and wrote C code to test the server connection using POST HTTP requests from the microcontroller. I also wrote documentation for the web app code and configured deployment (Vercel, environment variables) and kept the repository in sync between the UCSC GitLab and GitHub remotes.

In spring quarter, I mainly worked on the firmware and mobile application. On the firmware side, I implemented Bluetooth Low Energy provisioning, allowing the mobile app to connect to the smart base and send WiFi credentials and authentication tokens to the device over BLE. I also worked on WiFi setup and connection handling on the microcontroller. On the mobile side, I set up the React Native iOS app using Expo, configured Supabase for data storage, and implemented core screens including Settings, Groups, and related user flows for managing devices and group membership.

## B.3 Collaboration

Using Gitlab, tasks are self assigned on the issue board that are closely related to our division of labor. From our individual contributions, we have weekly meetings for everyone to catchup on what we did in person and online using discord. Whenever individuals tasks have overlap or dependency need we collaborate on the task. Tests plans/code are made in communication with the original maker of a task.

# C Test Plan & Results

## C.1 Overview

There are multiple sections of our design that require testing individually, then together as a whole. The following list represents the pathway we will take when testing our prototype segments.

- MCU Data Reporting
- MCU Web Posting
- Vercel Receiving and Displaying Data
- Notification System
- Phone App Integration
- Prototype Setup Out-Of-The-Box

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## MCU Data Reporting

The pressure sensors we use to build the base will output data to our MCU device. We want to be sure the MCU is capable of receiving this data and displaying it to the debug terminal. Once we can verify posting to the terminal we will move to the next stage of the data reporting testing.

With data being displayed properly, we can now use this to interpret what an empty versus full pitcher could be expected to weigh. The first test will include to test a known weight to make sure we get accurate weight readings. Next we will ensure that when there is no weight on the plate, the data will be read as 0g. Once this is correct, we will move on to the more complex calibration with Pitcher + water.

We will use different types of containers given the variety of containers possible to be used by customers. We will test at different set water levels, and verify that the expected weight represents the following formula.

$$W_{total} = W_{container} + V * \frac{g}{mL} \quad (1)$$

Water weighs 1 gram per milliliter so if we keep track of the volume we put into the container we know exactly what to expect for output weight if we know the base weight of the container alone.

## MCU Web Posting

To test this functionality the simplest way is to have the MCU send some sort of HTTPS POST pointed at one of our local machines. We can receive and print this data and then move forward to testing sensor data output.

With a working sensor we can trigger a POST request when an event happens. In this test case it could be picking up the container from the base which triggers a POST request with some set information. Upon successful receipt of this we can confirm that the HTTPS method is working on a small scale and move to the next segment of testing.

## Notification System

With data being sent out of the MCU, the next stage is to test if we can send a notification on some set event. For a test case this could be a container being placed on the device. The data sent as part of this notification could be a value representing the weight of the container, or an estimate of how much liquid is inside.

If we know how much liquid is inside the container when we place it on the base, it is easy to verify if the value we receive in the notification is the correct volume. Using different volumes we can create a table to show how close the reported volume was, and tweak our math to get it closer to accurate with different containers.

When we pass the fake data tests, we will compare the debug terminal and the notification water levels to ensure that it is correct.

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## Phone App Integration

With the notification system working, we can now test if the notifications are being sent to the phone app. We will use the same test case as before, which is placing a container on the base. We will verify that the notification is received on the phone and that the data in the notification is correct.

We will also verify that the current fill level of the container is being displayed on the phone app. We will test this by placing a container with a known volume of liquid on the base and verifying that the fill level displayed on the phone app is correct.

## Prototype Setup Out-Of-The-Box

In order to set up the prototype, the web app must connect to the MCU through BLE and scan a QR code to enter the WiFi credentials. To test this, we will try to connect to the MCU on boot multiple times using different devices and see if it is still able to connect. We will also test different types of WiFi networks, such as mobile hotspots, traditional password-protected networks, and networks with outside authorization such as university WiFi.

The user must also calibrate the device by measuring the empty and full weights of the filter on the weight sensor. On the mobile app, they will be given instructions step-by-step to do this. In order to test this process, we will give the device to a new user and see if they are able to clearly follow the instructions and calibrate the device correctly. We will also test robustness by intentionally disobeying the instructions, such as by weighing it empty both times, to see our device handle it.

# D Review

## D.1 Team Reflections

**Dev Chodavadiya:** I think the project went well overall. The team did a good job splitting up the work and communicating when things needed to come together. On my end, the 3D printing process took more iterations than expected to get the plates flat and properly aligned, but working through those issues taught me a lot about designing for manufacturability. I also worked on the feature of having the user signout everywhere, and as well deletion of the users account. If there was one thing I would do again, I would start the CAD and printing process earlier to leave more room for iteration. I think I would want to get more involved on the security side of this project, because I feel like that is an important area to improve my knowledge in.

**Logan Bossuwe:** The project went overall well coordinated roles. Collectively, we had a good idea of our needs and goals statement to work into a clear design path. We followed our planned schedule of implementations well and constantly kept communicating what was done. One thing we could have done better is to properly plan out an aesthetic prototype earlier and make it more flexible towards adding other additions to the circuit, such as a reset

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button. The current prototype does not have enough space and would need a reprint in order to design for a button/other features. What I could have done better is to start soldering the board containing the microcontroller and load cell amplifier earlier for our functional prototype. This would allow the rest of the team to have a clearer map of what was needed for the aesthetic prototype.

**Pieter Nauwelaerts:** The group worked really well given the time we had together. I think we had a clear idea of which sections we wanted to put work into, and overall everyone was able to get an idea of what was happening at each corner of the design. One thing I think we could have done differently is trying to get another microcontroller running the code so we could imagine what it would look like to have two devices connected to one user account, and test functionality like notifications that way as well.

**Reid Graham:** I think development went well. We were able to divide the work evenly between ourselves, individually work on our submodules, and combine them into the prototype. One thing I would have done differently would have been to conduct more real user tests, and get more feedback on the notification system and if users were less likely to leave it empty. I would also have liked to contribute more to the mobile application, but I did not have easy access to a Mac device.

**Sam Lai:** I think the project overall has went really well. We were able to efficiently work on the hardware and software sides to complete the project successfully. I really liked how the samrt base and mobile app turned out, I think the smart base looks sleek and functional and the mobile app looks clean and is easy to use. If I were to do it again, I would have liked to focus more on the authentication and security of our mobile app because it is really lackluster in comparison to industry standards right now. I think if I was given a little more time, I could have researched and implemented a more secure auth system for users.

**Shriyan Gote:** I think the project went well overall. Getting the BLE provisioning working end-to-end from the firmware side on the microcontroller all the way through to the React Native app on iOS was the most rewarding part, since it required both sides to work together correctly at the same time. If I were to do it again, I would start development earlier to leave more room for iteration, especially on the firmware side where debugging can be slow. I would also focus more on testing the BLE and WiFi provisioning flows under different conditions, and I would like to add a few additional features to the mobile app, such as customizable notification thresholds and more detailed device status views.